A Comparative Study of Classical and Quantum Randomness in Video Game Loot Systems

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Abstract— Our research is focused on an investigation of the effect that both classical pseudo-random (PRNGs) and simulated quantum random (QRNGs) systems have on generating loot in games. By looking at fairness of loot generated, equity of loot distributed, patterns of streaks, and satisfaction of players, we are attempting to determine how using quantum methods of randomness will impact the player's experience compared to classical methods.

Index Terms—Quantum Randomness, Classical Randomness, Video Game Loot, Player Satisfaction, Fairness

1 Introduction

In video games, loot generation is an important aspect of the overall player experience, especially within RPGs and adventure games. In these types of games, how loot (or items/resources) is distributed can greatly affect how players feel about their accomplishments, progress, and engagement with the game. Historically, loot has been generated using classical pseudo-random number generators (PRNGs), which create the illusion of randomness using deterministic algorithms. PRNGs are relatively efficient to use and are widely used in the video game community; however, they can occasionally create patterns where players may feel that loot drops are less than fair.

Simulated quantum randomness is based on the principles of quantum mechanics, such as superposition and entanglement; therefore, an outcome generated by quantum randomness would be unpredictable. Thus, this research will explore how simulated quantum randomness affects loot distribution and if this distribution offers a better, more consistent, and fairer alternative to existing systems based on classical distributions. Factors of interest include fairness of loot, streak patterns (number of times an item appears), and satisfaction of players. We will evaluate the effects of the simulated quantum random on the overall experience of the player.

2 Related Work

There have been some new studies conducted recently on using both classical pseudo-random number generators (PRNGs) and quantum random number generators (QRNGs) for many different types of applications, for example, gaming and scientific simulation.

Herrero-Collantes and Garcia-Escartin (2016) have outlined and explained the technologies which allow for the use of QRNGs, and how these technologies may be used in many different types of fields, such as cryptography or scientific research and simulation. One of the primary advantages of using QRNGs is that they provide a pure random number source due to the nature of quantum physics, i.e., utilizing quantum optical methods for creating pure and reliable random numbers. This is a major advantage of QRNGs over PRNGs that generate random numbers via deterministic methods and algorithms [1].

Likewise, Ghersi et al (2017) carried out an analysis comparing PRNGs with QRNGs when modelling with molecular Monte Carlo simulations. They found that there was a noticeable difference in the starting point of the quantum field versus the random number generator's area of the task given there were many quantum random starting points and that different types of quantum models gave different statistical results when generating random numbers. Therefore, using PRNGs generated from low-quality random number generator sources causes substantial deviations from traditional methods of simulating molecules due to the system having small molecular amounts of water or aqueous solution. Quantum random number generation provides consistent statistically and predictable numbers therefore demonstrates the value of true randomness as being generated by QRNGs when performing Monte Carlo simulations in a highly sensitive computational task [2].

The significance of examining quantum randomness when designing games and simulating them has been highlighted by these studies, as a result, it appears that there may be a potential application for quantum random number generators within the realm of video game loot generation systems to improve commonality among generated items (i.e., fairness).

3 Aim of the Experiment

The purpose of this experiment is to analyze the equality of loot distribution based on two types of randomness, the classical method of using pseudo-random algorithms (this would be called NRM), and simulated quantum randomness (QRDN). The goal is to determine if quantum randomness leads to more consistent loot drops and decreases the frequency of streaks/dry periods. Therefore, it is hoped that using QRDN will increase player perception of fairness and satisfaction with the loot system.

4 Experimental Method

This study uses a within-subject design where each person who took part in the study played under both conditions of randomness, specifically both NRM (traditional pseudo-randomness) and QRDN (simulated quantum) versions of the game. And all participants played under both systems, both randomly, during one session to eliminate any factors related to an individual's ability to play and personal preference.

4.1 Implementation of the Unity-Based Video Game

A desktop video game has been developed using the Unity game engine, where players combat enemies for loot. Loot will be categorized into types of rarity. Loot will have four types of rarity (common, rare, epic, and legendary).

The loot distribution system for the games will be developed by using the Unity libraries developed by Unity for game mechanics and creating written scripts in C#. The rate and type of loot drops will be randomly generated using a random number generator, and rates and types of loot drops will be defined by the method of the two types of randomness NRM and QRDN. The rates of the loot drops will be defined in the written scripts as will ensure that the loot distribution is consistent for the two types of randomness.

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Fig. 1: Screenshot of the Unity-based video game.

An interactive results board was created with the Unity User Interface system for the game as well. There would be a results board to show participants a detailed breakdown of their loot drops and performance during game play after they finished using the interactive results board. Key elements displayed include the number of loot items collected, the rarity ratings on those items, and whether they experienced long streaks or dry streaks during game play. This gave participants information to help them evaluate the fairness of the loot system and their satisfaction with their total game play.

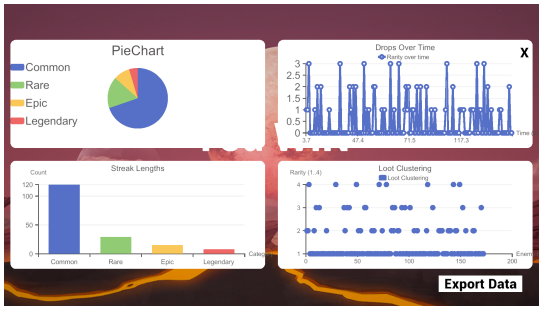


Fig. 2: Screenshot of the interactive results board.

4.2 Survey and Evaluation

After completing the game experience, participants completed an online survey measuring how they perceived fairness, how satisfied they were, and many other aspects of their overall game experience. The survey was comprised of Likert scale questions intended to gather subjective experiences and perceptions people had about the loot fairness, and consistency while playing.

5 Procedure

The recruitment of participants came from the Brigham Young University–Hawaii student body through class announcements and social media. Each participant completed the game by eliminating enemies as often as they could in both randomization (NRM and QRDN). The order in which conditions occurred was counterbalanced between participants for any potential effects of order on performance.

Instructions were distributed to all players prior to the start of the game. The purpose of these instruction manuals was to inform participants about how they could move throughout the game and how they could use the results board once they completed their game. Players were then informed that they would participate in two separate games utilizing loot drop schemas. Players were not provided any information as to which loot drop schema was being used at any time during the game play.

As each group of players finished playing their games, they were shown two sets of graphs representing how their loot dropped throughout their games. The graphs included one for loot distribution (a pie chart), one for the length of the player's streaks (a histogram), one for loot drops over time (a line chart), and one for the level of clustering or obtaining rare loot through time (a scatter plot). Participants then completed an evaluation of how they viewed the fairness, satisfaction level, and overall experience they received in playing under each random event.

6 Data and Analysis

The data collected in this experiment includes both quantitative and qualitative measures. The main components of the data are as follows:

1. **Loot Drop Data**: This includes the type and frequency of loot drops (Common, Rare, Epic, Legendary) for each randomness condition (NRM and QRDN). We focus on the total number of loot items dropped and the distribution of these items across different rarity categories.
2. **Survey Data**: Participants provided feedback on their perceptions of fairness, loot satisfaction, and their experience with streaks and dry spells in the loot system. The survey included Likert scale questions to capture their subjective experiences.

Statistical Analysis A variety of statistical analyses were done in order to evaluate randomness in terms of the following variables: random distribution of loot and how players perceive their random distributions.

1. **Loot Distribution Analysis**:
   (a) To begin, we evaluated whether the distributions of loot (e.g., number of Loot Items) were normally distributed between the two groups by conducting the Shapiro-Wilk test.
   (b) If both distributions were normally distributed, we would use the paired t-test to compare the means (e.g., average number of items) between the two groups (i.e., normal vs non-normal).
   (c) If neither distribution is normally distributed, we would use the Wilcoxon signed-rank test (i.e., non-parametric alternative) to compare both distributions.
2. **Survey Data Analysis**:
   (a) Participants' answers to each survey question were coded with numbers, and descriptive statistics (mean, median, and standard deviation) were computed to summarize the players' perceptions of fairness, satisfaction, and the distribution of the number of consecutive games played.
   (b) To make comparisons between player perceptions of loot fairness and satisfaction between NRM and QRDN, we used the Mann-Whitney U test to compare the two independent player groups when their distribution is not normal.

7 Assumptions & Justifications

The statistical tests selected for this study were chosen according to the type of data and the assumptions used in our analysis. We selected the tests to provide an appropriate basis for comparing loot distributions and player's perceptions of classic pseudo-randomness (NRM) to quantum randomness (QRDN).

1. **Normality of Loot Distribution**: A Shapiro-Wilk test was performed on the loot distribution data to determine if it followed a normal distribution. This is a valid test for small sample sizes and is considered to be a good way to evaluate normal distribution.
2. **Fairness and Player Perceptions**: Player perception data (Likert scale responses) was ordinal in nature so we used non-parametric tests, such as the Mann-Whitney U test, to compare player perceptions of fairness, streaks, and satisfaction between NRM and QRDN conditions. Non-parametric tests are appropriate for comparing data that do not follow a normal distribution.

Expectations: We hypothesized that players would have a more positive perception of classical pseudo-randomness (NRM) in regards to fairness and consistency compared to quantum randomness (QRDN),

which we anticipated to provide a more consistent loot drop experience and ultimately enhance the gaming experience in terms of both fairness and satisfaction for players.

8 Results of Statistical Tests

Our analysis produced the following results:

1. **Loot Distribution Analysis**:
   (a) The loot distribution datasets for Common, Rare, Epic, and Legendary loot for each randomness type were normally distributed, as determined by the Shapiro-Wilk test (p-values for NRM = 0.1494; QRDN = 0.1565), therefore normality can be established.
   (b) Using a paired t-test, we compared the loot distributions of each randomness type. There was no statistically significant difference between the two types of randomness ($p = 1.0$), indicating that randomness type does not appear to significantly affect the overall loot distribution of Common, Rare, Epic, and Legendary loot. Nevertheless, the distribution patterns of Rare and Epic loot drops seemed to be more homogenous under QRDN, however this observation reached no significance statistically.
2. **Player Perception Analysis**:
   (a) We used the Mann-Whitney U Test to gauge the fairness and satisfaction of players via survey. There were no statistical differences between the fairness scores provided by players who used either method of randomness, as indicated by the p value of 0.2632. However, there appears to be a trend of players indicating that they perceive the classic form of randomness (NRM) to be slightly more fair than the quantum variety (QRDN), as was indicated by the results of the Mann-Whitney U test.
   (b) Additionally, the analysis of players' perception of streaks or periods of inactivity did not yield statistical significance utilizing either method of randomness, as indicated by the p value (> 0.05). Furthermore, while there were no statistically significant results, trends that indicate QRDN may have provided players with more enjoyable or uniform experiences related to streaks and dry spells could benefit from further investigation in the future.

In conclusion, trends have been established suggesting that more research should be done to explore how players perceive fairness in loot distributions. The players in the study tended to see classical randomness (NRM) as marginally fairer than quantum randomness (QRDN), although this was not statistically significant. Trends in rare and epic loot drops also indicated that QRDN provided more consistency than NRM in these areas. However, there was no evidence that either type of randomness affected players' perception of streaks or dry spells, though QRDN may provide a more consistent experience for players in these areas. These trends may indicate that QRDN might provide possibilities for further research in enhancing player game experiences.

9 Results/Findings

9.1 Loot Distribution Analysis

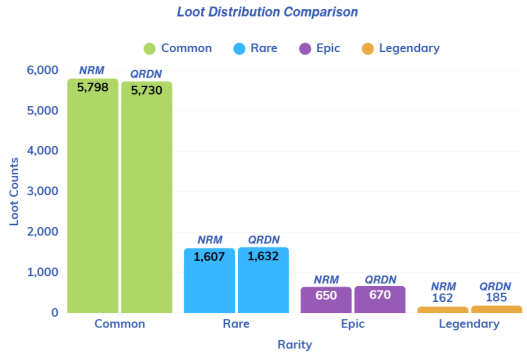


Fig. 3: Loot distribution comparison between classical (NRM) and quantum (QRDN) randomness.

The data collected on the loot distribution (Common, Rare, Epic, Legendary) for both Randomness Modes of the Game were tested using the Shapiro-Wilk test for normality. In Figure 3, it can be seen that both Randomness Modes of the Game resulted in normally distributed data, as both showed statistical normality using the Shapiro Wilk test; this means there is no substantial difference in how players received their loot regardless of which type of Randomness Mode was used.

9.2 Streak Length Analysis

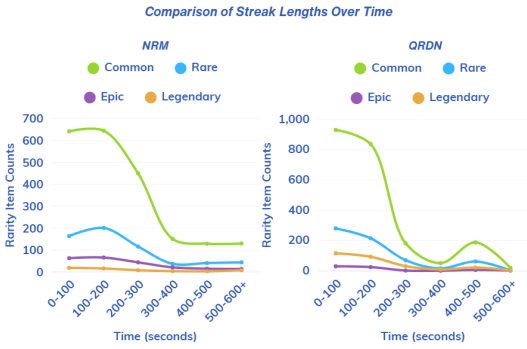


Fig. 4: Comparison of streak lengths over time for classical (NRM) and quantum (QRDN) randomness.

The loot rarity comparison, as shown in Figure 4 (Common, Rare, Epic, Legendary), was made based on streak lengths over time. From this analysis of the streak length data, no statistical difference was found in loot streak length based on Randomness Mode, meaning that both Randomness Modes produced the same streak duration.

9.3 Player Perceptions

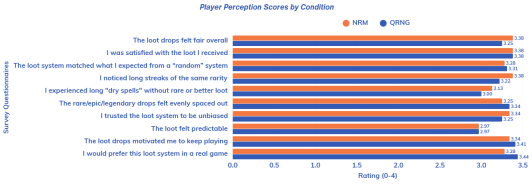


Fig. 5: Player perception scores comparing fairness, satisfaction, and streak patterns between NRM and QRDN.

From the player survey data shown in Figure 5, it was found that player perceptions of fairness and satisfaction were also similar across both Randomness Modes, indicating that the method of producing the loot did not have a significant effect on the overall gaming experience.

10 Conclusion

There was no significant difference between classical pseudo-randomness and simulated quantum randomness in both loot distributions and players' perceptions of fairness. While there were some expectations based on prior research that players would experience a greater level of loot harmony when using quantum random numbers, the test results indicate an insignificant disparity of the level of satisfaction players felt towards the fairness of loot based upon the type of method used to generate the loot. Therefore, this research establishes a starting point for additional investigations into the possible implementation of quantum random number generation in future game development processes.

11 Discussion

The findings from this study indicate that while quantum randomness theoretically can provide true randomness, there was no appreciable difference in the distribution of loot, or in player perceptions of fairness and satisfaction, when using pseudo-randomness.

One reason for this may be due to the fact that the random generation samples in this study were theoretically generated rather than actually created by actual quantum mechanics and therefore may not have demonstrated the theoretical advantages of true quantum randomness. In the real world, quantum randomness may provide a greater difference than in this study, particularly when applied to larger and more complex systems, where small variations between these two methods would accumulate over time. It should also be noted that due to the relatively short duration of play in this experiment and only a limited number of loot drops (i.e., there were only eight different types of loot) it is likely that no difference between these two methods would have been discernible.

Findings from this study also show that players' perceptions regarding streaks and/or dry streaks was not significantly affected by whether the loot was randomly awarded through true chances, video game mechanics, or manipulation (e.g., player's record). This would indicate that other things—like the design of the video game, player's expectations, and their overall experience while playing—may be greater influences on players' perceptions regarding fairness and satisfaction with loot drops than the actual way in which they are awarded.

11.1 Future Work

There are a number of limitations in this study and these should be addressed in the future. First, there were a limited number of participants, 45 students. The data would be better with more participants from a wider range of backgrounds. Future studies should try to gather data about a wider range of people, which would lead to more accurate and generalizable results.

Secondly, the respondents' knowledge and comprehension of both the game they were playing and the survey questions may have affected their answers to the questions they were asked. In the study, the players made their own analyses of fairness, satisfaction, and randomness, which could have introduced additional bias as well. Future research should have provided clearer instructions as to what to focus on when playing the game and provided guidance on how the game's experience would differ from other games to help ensure that all respondents' answers were measured against the same set of criteria; thus providing a basis for more valid and less biased data.

In addition, future studies should examine different types of quantum random number generation. For example, instead of using random number generators that are software-based (which are typically considered not to be true forms of randomness), researchers should use hardware-based generators to determine whether a true quantum random number will provide a greater advantage to game design than a non-quantum form of random number generation. To enhance the existing understanding of quantum random numbers, researchers should investigate their effects on larger, more complex gaming atmospheres by using longer gameplay periods and more loot dropped. By administering and assessing both quantum and non-quantum forms of random numbers during longer game sessions with more loot dropped will give players greater benefits than simply incorporating only one type of random number in the design of the game.

One additional area to explore is how people perceive randomness based on the type of game they play (i.e., are role playing games really dependent on random loot drops for progression as opposed to say first person shooters where loot is probably less impactful). This might give us better insights into the ways randomness has an effect on games and gamers of each type.

Ultimately, the findings from this study have provided some insight into whether or not utilizing quantum randomness will provide an advantage over classical pseudo-random methods for video game designs; and therefore further research into the context where the increased fairness and consistence from utilizing the properties of quantum randomness can improve the player's gaming experience is needed.

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References

[1] M. Herrero-Collantes and J. C. Garcia-Escartin, "Quantum random number generators," *arXiv:1604.03304v2*, 2016. [Online]. Available: https://arxiv.org/abs/1604.03304v2 1

[2] D. Ghersi, A. Parakh, and M. Mezei, "Comparison of a quantum random number generator with pseudorandom number generators for their use in molecular monte carlo simulations," *Journal of Computational Chemistry*, vol. 38, no. 31, pp. 2713–2720, 2017. 1